

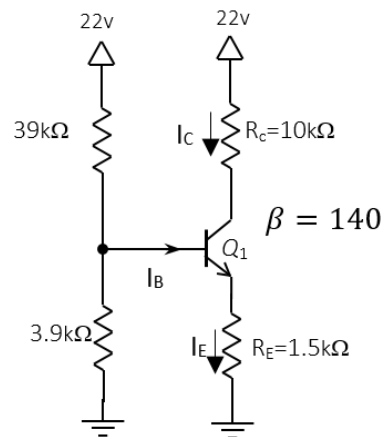
ASSIGNMENT-5

Full Marks 40

Q1. [CO2]

10

Analyze the circuit to determine V_{CE} and I_C . You must validate your assumption.

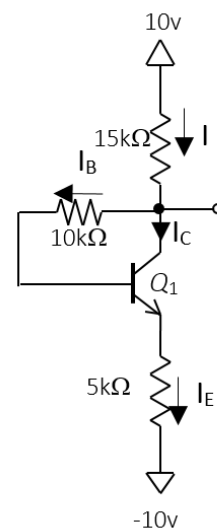


Q2. [CO2]

10

Assume Q1 is in forward active region. **Analyze** the circuit to find I_C and V_C . Assume $\beta = 100$.

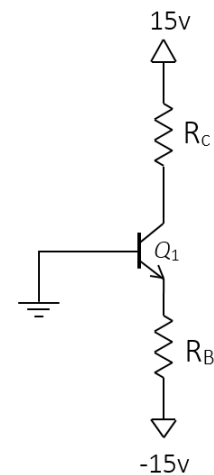
[Hints $I = I_E$]



Q3. [C03]

10

Design the circuit so that a current of 2 mA flows through the collection in Forward active operating mode and a voltage of +5 V appears at the collector. Assume $\beta = 100$.



Q4. [C02]

10

For the transistor $\beta = 75$. **Analyze** the circuit to find the Values of R_C , all node voltages and branch currents.

